HATELENS: Hateful Video Detection with Cues from Dual Perspectives

Anonymous ACL submission

Abstract

Hateful video detection poses greater challenges to the online ecosystem, as hateful content in video form spreads more rapidly and can have a stronger impact than traditional forms of dissemination. Existing detection methods fall short in understanding how hateful meaning is formed in videos and struggle to handle ambiguous cases involving implicit hate. In this work, we propose HATELENS, a dual-perspective paradigm that integrates cues from both internal and external environments. Internally, we leverage multimodal large language models (MLLMs) through a perception-to-cognition reasoning process, producing structured meaning formation for understanding hateful context in videos. Externally, we propose a test-time retrieval and adaptation mechanism that operates on fused multimodal representations to exploit annotated precedents, providing richer local evidence to help disambiguate implicit hate. Extensive experiments on three public datasets demonstrate that our method achieves state-of-the-art performance, with macro-precision improvements of up to 16%. Code of our project is available at...

Disclaimer: This paper contains content disturbing to some readers.

1 Introduction

The proliferation of short-form video platforms such as YouTube Shorts, TikTok, and Bilibili has created new vectors for the dissemination of hateful content (Das et al., 2023; Wang et al., 2024a). Unlike text-only hate speech, hateful videos embed harmful messages across multiple modalities—visual imagery, spoken language, on-screen text, and audio cues—making them particularly challenging to detect and moderate (Kiela et al., 2020; Lang et al., 2025). The multimodal nature of video hate demands methods that can reason across these diverse signals rather than merely aggregate them.

Existing approaches to hateful video detection fall into two categories. *Multimodal feature fusion methods* extract representations from each modality using pre-trained encoders (e.g., BERT for text, ViT for frames, audio models for speech) and combine them through attention mechanisms (Das et al., 2023; Céspedes-Sarrias et al., 2025), mixture-of-experts architectures (Lang et al., 2025), or graph-based fusion (Yue et al., 2025). On the other hand, the rise of multimodal large language models (MLLMs) proves strong reasoning capability that can help both explainability and accuracy of detecting hate videos. Therefore, *MLLM-based methods* leverage MLLMs to analyze video content, typically through chain-of-thought (CoT) prompting that first describes visual and textual elements, then combines these descriptions into a judgment (Jing et al., 2025; Yang et al., 2025).

However, two important challenges remain underexplored. **(C1) Lack of structured meaning formation for hateful video understanding.** First, multimodal feature fusion methods model *what* is present in the video, but not *how* the content should be interpreted. This creates a dilemma in which the model learns surface-level features without learning to contextualize why the content is hateful, thereby limiting the generalizability of the learned representations. Although the rise of MLLMs has created new opportunities to address this issue, current MLLM-based methods (Jing et al., 2025; Yang et al., 2025, 2026) still remain at the level of perceiving and analyzing surface features, or they begin with a predefined hateful/non-hateful judgment and then prompt the MLLM to search for supporting evidence. This introduces a clear prior bias, making the reasoning process less reliable. **(C2) Lack of effective ambiguity resolution for hateful video decisions.** Even when models learn to some extent *how* hate is expressed in video during training, the resulting representations are still likely to only support coarse-grained global deci-

sions that mainly cover common patterns. However, hateful content is often expressed in implicit and diverse ways to evade detection, making such cases inherently ambiguous and placing them near the decision boundary. Although some prior work (Lang et al., 2025) has introduced external retrieval to alleviate this issue, these methods typically operate on early multimodal features with manually specified modality weights, making them less effective for handling the dynamic nature of the video ecosystem.

Brunswik's Lens Model implies that judgments are formed by integrating cues from both internal and external environments. Inspired by this, we propose HATELENS, which shifts the paradigm of hateful video detection from a direct perception to discriminative classification problem to a process of internal reasoning with structured meaning formation, followed by ambiguity resolution through external precedent case consultation. Specifically, the *P2C Generator* first leverages the MLLM to organize multimodal observations of a given video into cognitive evidence through perception-to-cognition deduction, producing a structured reasoning schema for meaning formation, including *what* the hateful content is, *where* it is located, *why* it should be classified as hateful, and *how* it may promote hateful activity, thereby deeply contextualizing the video. The *Schema-guided Modality Router* then jointly fuses all multimodal features with the above textual reasoning schema to produce more discriminative feature representations. Finally, to address ambiguity, the *Whitened Neighbor Interpolator*, a gradient-free module, refines decisions through a test-time retrieval and adaptation mechanism that consults the nearest k annotated neighbors from the training set to provide local evidence. Unlike previous retrieval methods, it operates directly on fused multimodal representations that are further geometry-conditioned, ensuring that the retrieved neighbors are more closely aligned with the final decision semantics rather than merely exhibiting surface-level similarity. Our contributions are summarized as:

- We propose a new paradigm for hateful video detection, formulating the task as evidence-grounded decision formation from both internal and external perspectives.
- We introduce the *P2C Generator*, which uses an MLLM to transform internal multimodal observations into structured cognitive evidence through perception-to-cognition reasoning.
- We propose the *Whitened Neighbor Interpolator*, a geometry-conditioned test-time retrieval module that refines ambiguous decisions with external annotated precedents that are more closely aligned with the final prediction space.

We conduct comprehensive experiments on three benchmarks across two languages, achieving state-of-the-art results with detailed analysis of both reasoning quality and retrieval effectiveness.

2 Method

Problem Formulation. Let q denote a video on online platforms. Each video q is represented by its multimodal content, including textual, visual, and audio modalities. For each video, we extract textual, visual, and audio information. For text, we directly use the video's metadata t, including title and video description. For visual content, we uniformly sample k frames from the video, denoted as $v = \{v_1, .., v_k\}$. For audio a, we obtain a global transcript t_a using automatic speech recognition (ASR). The task is to determine whether the video q contains hateful content by jointly modeling all modalities t, v, a.

Overview. The overall framework of our method is illustrated in Fig. 1. It follows the general process of human cognitive reasoning from both internal and external perspectives, progressing from perception to cognition to response, and is instantiated through three modules: (1) the *P2C Generator*, which employs the MLLM to internally examine the given video in depth, moving from global and local perception of the content to summarized cognitive evidence; (2) *Schema-guided Modality Routing*, a fusion module that models the interaction between multimodal video features and the cognitive schematic evidence produced by the MLLM in the latent space; and (3) the *Whitened Neighbor Interpolator*, which adopts a gradient-free test-time retrieval mechanism to externally identify precedent nearest neighbors with annotated ground truth and refine decisions on detected samples with reduced ambiguity.

2.1 P2C Generator

Most multimodal fusion-based methods directly map observable features to final judgments, skipping the crucial intermediate step of cognition.

[PLACEHOLDER: Framework Overview Figure]
Module 1: CAT-Guided Dual-Stage LLM Analysis → Module 2: Appraisal-Conditioned Fusion → Module 3: Case-Based Retrieval-Augmented Inference

Figure 1: Overview of the HATELENS framework. Given a short video, (1) the CAT-guided dual-stage LLM pipeline first extracts an evidence ledger from multimodal content, then performs structured appraisal inference; (2) the appraisal-conditioned fusion module integrates raw media features with LLM-derived appraisal representations; (3) at inference time, the case-based retrieval module refines uncertain predictions by consulting similar labeled instances in a geometrically conditioned embedding space.

While some MLLM-based methods incorporate additional textual analysis, they still fail to fully unlock the reasoning capabilities of MLLMs, remaining largely at the surface level. Ideally, the modeling approach should mimic human annotators, who follow a complete process from perception to cognition to evidence-supported decision—consistent with cognitive appraisal theory. To this end, we propose a *P2C Generator* that incorporates a perception-to-cognition CoT to further exploit MLLM reasoning capabilities. As shown in Fig. 2, the MLLM is prompted to generate `<think>...</think>` and `<answer>...</answer>` tags, where it first progresses from perception to cognition and then summarizes all identified evidence.

Stage 1: Perception. Given a video t, t_a, v, a, humans first observe the video holistically. When signals of hateful content appear, they allocate more localized and focused attention to the relevant clips. Following this process, in Stage 1, which consists of Step 1 and Step 2, we expect the model to first form a global perception through *Scene and Context Description*, which provides an objective overview of the environment, participants, visible text, symbols, and overall activity in the video, and then conduct localized perception through *Hateful Cues Description*, which describes specific verbal, textual, visual, or symbolic content that may indicate hateful expression.

Stage 2: Cognition. After acquiring sufficient surface-level information, humans conduct reasoning based on the perceived cues and infer possible consequences. In hateful video assessment, this process moves from shallow to deep understanding: it first identifies who or what is being targeted, if anyone, and explains the type of attack involved, such as insult, harassment, or humiliation, which we term *Target and Attack Identification* (Step 3); it then performs *Contextual Harm and Norm Reasoning* (Step 4), which explains why the content should be considered hateful in context.

Stage 3: Answer. After completing the reasoning process, the MLLM is expected to provide a short structured summary related to the key evidential dimensions of hateful content in the video, including the classification (**which**), a description of the hateful content (**what**, w), the spatio-temporal-audio key evidence (**where**, e), the reason (**why**, r) for this classification, and **how** (o) it may promote hateful activity and cause harm.

Note that all three stages are completed within a single prompt call. In the subsequent modules, we use only the structurally defined fields marked by the mathematical symbols from the returned answer, as they provide a concise summary of the preceding thinking process while preserving all the key evidence we require.

2.2 Schema-Guided Modality Router

Modality routing has been shown to be an effective fusion paradigm for multimodal feature integration. In this regard, we view the structural fields returned by the MLLM (i.e., w, e, r, and o) as a reasoning schema that can more effectively guide multimodal fusion.

We map each modality input into latent features using the corresponding off-the-shelf encoders, including a pre-trained language model (LM), visual transformer (VT), and audio transformer (AT), as:

$$\begin{aligned} \boldsymbol{h}_t &= \mathrm{LM}\left(t; t_a\right), \\ \boldsymbol{h}_v &= \mathrm{VT}(v), \\ \boldsymbol{h}_a &= \mathrm{AT}(a), \end{aligned} \tag{1}$$

where $\boldsymbol{h}_* \in \mathbb{R}^d$ denotes the extracted feature vector after applying mean pooling, and ; denotes concatenation. Accordingly, we also encode the textual structural fields separately using the same LM, resulting in $\boldsymbol{h}_{w/e/r/o}$.

Subsequently, we fuse the seven resulting feature vectors using a multi-head gated routing mechanism:

$$\begin{aligned} \alpha_m^{(j)} &= \frac{\exp(f_j(\mathbf{h}_m))}{\sum_{m' \in \mathcal{M}} \exp(f_j(\mathbf{h}_{m'}))}, \\ \boldsymbol{g}^{(j)} &= \sum_{m \in \mathcal{M}} \alpha_m^{(j)} \boldsymbol{h}_m, \end{aligned} \tag{2}$$

where $\mathcal{M} = \{t, v, a, w, e, r, o\}$. The final representation is formed by concatenating the outputs of all routing heads with a mean-pooled modality summary, and is then passed through a classification MLP:

$$\begin{aligned} \bar{\boldsymbol{g}} &= \mathrm{MLP}([\boldsymbol{g}^{(1)}; \ldots; \boldsymbol{g}^{(H)}]) \\ p_{cls} &= \mathrm{MLP}\left(\bar{\boldsymbol{g}}\right), \end{aligned} \tag{3}$$

where $\bar{\boldsymbol{g}}$ is the fused representation and $p_{cls} \in \mathbb{R}^2$ denotes the predicted probability distribution over the hateful and normal classes for the input video. The model is trained with the cross-entropy loss.

2.3 Whitened Neighbor Interpolator

Hate judgments are often inherently ambiguous: the same surface content may be hateful, satirical, educational, or counter-speech depending on context and framing. In the video setting, such expressions are even more diverse yet individually rare, resulting in a long-tail distribution. To address this, we propose leveraging training samples with similar fused features and annotated ground truth to sharpen the decision boundary. This design mirrors content moderation practice, where moderators learn general policies but consult prior precedent cases for difficult judgments (Klonick, 2018; Clarke et al., 2023). Specifically, we split this into two steps.

Geometric Conditioning. Previous methods (Lang et al., 2025) for multimodal retrieval typically operate directly on the initially extracted multimodal features with manually configured modality weights. This is problematic for two reasons. First, such designs are difficult to adapt to dynamically varying cases. Second, the initial features are not sufficiently discriminative for the final classification objective, which limits the usefulness of the retrieved cases for informing decisions. This motivates us to instead leverage the fused penultimate representations $\bar{\boldsymbol{g}}$. However, these representations combine heterogeneous sources, including raw multimodal features and LLM-derived reasoning, which results in an anisotropic embedding space where distances become unreliable (Yi et al., 2025; Tsukagoshi and Sasano, 2025). To enable meaningful similarity-based retrieval, we apply covariance-based whitening:

$$\hat{\boldsymbol{g}} = \frac{(\bar{\boldsymbol{g}} - \boldsymbol{\mu})\mathbf{W}_{\text{whiten}}}{\|(\bar{\boldsymbol{g}} - \boldsymbol{\mu})\mathbf{W}_{\text{whiten}}\|_2}, \tag{4}$$

where $\boldsymbol{\mu}$ is the training set mean and $\mathbf{W}_{\text{whiten}}$ is derived from shrinkage-regularized covariance estimation (Yi et al., 2025). This conditioning ensures that similarity computations reflect genuine semantic proximity rather than artifacts of the heterogeneous feature geometry.

Retrieval and Interpolation. At inference, we store whitened penultimate representations $\{\hat{\boldsymbol{h}}_i\}_{i=1}^N$ from the training set along with their labels. For a test sample with representation $\hat{\boldsymbol{h}}_{\text{test}}$, we retrieve its k nearest neighbors and compute a soft label distribution:

$$p_{\text{kNN}}(y = c) = \frac{\sum_{i \in \mathcal{N}_k} \mathbb{1}[y_i = c] \cdot \kappa(\hat{\boldsymbol{g}}_{\text{test}}, \hat{\boldsymbol{g}}_i)}{\sum_{i \in \mathcal{N}_k} \kappa(\hat{\boldsymbol{g}}_{\text{test}}, \hat{\boldsymbol{g}}_i)}, \tag{5}$$

where $\mathcal{N}_k$ denotes the k nearest neighbors and $\kappa(\cdot, \cdot) = \exp(\mathrm{sim}(\cdot, \cdot))$ is a similarity kernel, which enhances weights from more similar neighbors. The final prediction interpolates the classifier logits with the retrieval distribution:

$$p_{\text{final}} = (1 - \alpha) \cdot p_{cls} + \alpha \cdot p_{\text{kNN}}, \tag{6}$$

where α is the trade-off parameter.

Table 1: Data statistics of three benchmarks.

Dataset	#Vid	Split			Class Dist.		
		Trn	Val	Tst	Nor.	Off.	Hat.
HateMM	1,066	744	107	215	639	–	427
MHClip-Y	891	618	91	182	601	218	72
MHClip-B	897	633	88	176	605	180	112

3 Experiments

3.1 Experimental Setup

Datasets. We evaluate our method on three hateful video detection benchmarks: HateMM (Das et al., 2023), MultiHateClip-Youtube (MHClip-Y), and MultiHateClip-Bilibili (MHClip-B). We adopt the official data splits from Lang et al. (2025), and Table 1 summarizes the dataset statistics. Following prior work, MHClip-Y and MHClip-B are evaluated under a binary setting (Normal vs. Offensive+Hateful). Results for three-way classification are provided in the Appendix.

Baselines. We compare against baselines from three categories: (1) **Unimodal**: BERT (Devlin et al., 2019), ViViT (Dosovitskiy et al., 2021), MFCC (Davis and Mermelstein, 1980); (2) **Multimodal fusion**: Pro-Cap (Cao et al., 2023), HateMM (Das et al., 2023), MHCL (Wang et al., 2024a), MoRE (Lang et al., 2025), ImpliHateVid (Rehman et al., 2025), CMFusion (Zhang et al., 2024); (3) **MLLM-based**: MiniCPM-V, LLaVA-OV (Li et al., 2024), Qwen2-VL (Wang et al., 2024b), HVGuard (Jing et al., 2025). Details of baselines are in Appendix.

Metrics. Following previous work (Lang et al., 2025; Jing et al., 2025), we report Accuracy (**ACC**), Macro-F1 (**M-F1**), Macro-Precision (**M-P**), and Macro-Recall (**M-R**). The best checkpoint selected on the validation set is then evaluated on the test set for performance reporting. All results are averaged over 20 random seeds.

Implementation Details. We use `GPT-5.4-nano` as the MLLM backbone. Text features are encoded with `BERT` (Devlin et al., 2019), visual features with `ViT` (Dosovitskiy et al., 2021), and audio features with `WavLM` (Chen et al., 2022); all encoders are kept frozen. The fusion module uses $H = 4$ routing heads with hidden dimension $h = 192$. For retrieval, we set $k = 15$ and $\alpha = 0.2$. Further implementation details are provided in Appendix A.

3.2 Comparison with SOTA

Table 2 presents the main results, from which we highlight three key observations. **(O1): Multimodal fusion substantially outperforms unimodal approaches**, as hateful expression may emerge across different modalities that single-modality models cannot capture. **(O2): Structured MLLM reasoning provides substantial gains over surface-level fusion.** MLLM-based methods (e.g., HVGuard) consistently outperform feature-fusion baselines (MoRE, MHCL, ImpliHateVid), and our method further extends this advantage by guiding the MLLM through a structured perception-to-cognition process that explicitly models evidence, targets, and contextual harm. **(O3): HateLens achieves the strongest performance across all three datasets on all metrics.** Compared with HVGuard, HateLens yields absolute accuracy gains of 3.7%, 5.5%, and 9.6%, respectively, while its higher precision and recall indicate more hateful content detected with fewer false positives. These results validate our paradigm of first examining videos through progressively deeper perception and cognition, then consulting external cases to support the final decision.

3.3 Ablation Study

Table 3 reports the ablation results across three benchmarks. **Effect of P2C Generator.** Removing individual answer fields (*-what*, *-where*, *-why*, *-how*) consistently degrades performance, with *what* and *how* showing the largest impact, indicating that both content description and harm contextualization are essential. Using only the perception stage (Percep. only) or only the cognition stage (Cogn. only) underperforms the full answer, confirming their complementary roles. Replacing our P2C prompt with the HVGuard CoT prompt (Jing et al., 2025) while keeping the same fusion and retrieval modules causes a substantial drop, demonstrating that structured perception-to-cognition decomposition produces more informative representations than a describe-analyze-judge pipeline. **Effect of Schema-Guided Router.** Substituting our fusion module with those from MoRE and HVGuard leads to notable degradation, which we attribute partly to the optimization difficulty and expert collapse risks of MoE-based architectures, whereas our router is more stable and effectively integrates multimodal features with LLM-derived reasoning. **Effect of Whitened Neighbor Interpolator.** Removing the

Table 2: Comparison with state-of-the-art methods on three hateful video detection benchmarks. Best results in **bold**, second best underlined. ↑ indicates the improvement of HATELENS over the best baseline.

	HateMM				MHClip-Y				MHClip-B			
Method	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R	ACC	M-F1	M-P	M-R
Unimodal Methods												
BERT	69.1	63.7	70.1	64.0	65.5	49.1	55.2	52.2	72.5	67.7	68.4	62.8
ViViT	68.2	66.7	66.8	66.6	61.4	61.4	62.2	61.1	71.0	66.1	66.6	65.8
MFCC	65.4	60.3	64.1	60.7	46.5	47.2	58.8	52.2	63.1	52.5	54.1	53.0
Multimodal Fusion Methods												
Pro-Cap	64.5	63.3	63.4	63.2	70.1	66.3	66.3	66.3	72.5	66.8	66.1	68.3
HateMM†	76.0	72.8	77.9	72.0	71.5	63.2	68.3	62.6	71.0	61.8	66.5	61.4
MHCL	77.4	76.5	77.5	76.6	65.4	65.4	67.2	64.9	76.5	73.1	73.2	73.0
MoRE	83.4	82.4	81.8	83.3	77.5	75.2	75.7	74.8	78.5	74.8	75.7	74.1
ImpliHateVid⋆	82.8	82.0	82.1	82.0	76.1	71.7	71.6	71.8	78.3	72.8	74.4	71.8
CMFusion	75.6	74.1	75.3	73.7	72.2	68.8	69.5	68.2	73.8	70.1	71.0	69.4
Large Vision-Language Models												
MiniCPM-V	72.4	72.3	77.8	76.4	69.1	67.4	69.3	67.4	73.5	73.2	72.0	73.0
LLaVA-OV	75.6	75.6	77.9	78.3	73.7	70.5	70.5	66.8	75.2	70.2	71.4	70.3
Qwen2-VL	73.7	73.7	78.1	77.3	70.5	66.8	66.8	66.7	73.3	74.0	73.9	72.9
HVGuard⋆	87.9	87.5	87.3	87.6	79.8	76.2	77.0	75.6	79.6	75.6	76.0	75.3
HATELENS	**91.6**	**91.4**	**91.0**	**91.9**	**85.3**	**80.5**	**86.3**	**77.8**	**89.2**	**85.6**	**92.0**	**82.5**
↑ *Improv.*	*+3.7*	*+3.9*	*+3.7*	*+4.3*	*+5.5*	*+4.3*	*+9.3*	*+2.2*	*+9.6*	*+10.0*	*+16.0*	*+7.2*

⋆ Reproduced by us on the same data split. All numbers are percentages (%).

Table 3: Ablation study (%). Avg. Δ: mean ACC drop across three datasets.

	HateMM		MH-Y		MH-B		
	ACC	M-F1	ACC	M-F1	ACC	M-F1	$\overline{\Delta}$
Full	**88.4**	**87.7**	**83.4**	**78.8**	**89.2**	**85.6**	—
P2C Generator							
–what	86.9	86.4	82.1	77.0	86.6	83.0	–1.8
–where	87.9	87.3	82.8	77.8	84.7	83.1	–1.6
–why	87.6	87.0	82.2	77.4	85.4	84.0	–1.6
–how	87.1	86.2	81.9	76.6	84.7	82.2	–2.7
Percep. only	86.2	85.2	80.9	75.1	83.5	80.9	–4.1
Cogn. only	85.6	85.0	80.2	74.3	82.1	84.1	–3.5
HVGuard CoT	85.1	84.3	79.1	73.2	78.3	73.2	–6.8
Schema-Guided Router							
MoE fusion	83.7	82.9	77.9	70.7	81.5	79.1	–6.7
HVGuard fus.	87.0	86.1	78.5	74.0	79.0	76.6	–5.2
Whitened Neighbor Interp.							
–Interp.	87.0	86.4	80.4	75.4	86.0	82.9	–2.7
–whitening	88.4	87.7	79.1	74.3	85.4	82.2	–2.3
Pre-fus. retr.	85.6	84.4	82.8	77.2	82.2	73.7	–3.7

interpolator (-Interp.) causes a clear drop, confirming the value of consulting external cases. Geometric conditioning (-whitening) is also important, as the fused representations are anisotropic and less reliable without this step. Retrieval based on initial multimodal features (Pre-fus. retr.) also underperforms, indicating that such a mechanism struggles with the complex and dynamic nature of this task.

3.4 P2C CoT Analysis

An effective MLLM-based CoT should exhibit two properties: (1) the output reasoning should provide informative evidence that aids human decision-making, and (2) the CoT design should generalize across different MLLMs with consistent performance gains. We evaluate both through quantitative analysis, comparing against HVGuard (Jing et al., 2025) and RAMF (Yang et al., 2025).

Explanation Quality. We randomly sample 100 videos of each dataset and evaluate the reasoning outputs from three methods using GPT-4o as an expert judge across five criteria: (1) *Informativeness*: whether the explanation provides background and context beyond surface description; (2) *Soundness*: whether the reasoning is valid and logically coherent; (3) *Persuasiveness*: whether the conclusion is convincing; (4) *Readability*: whether the text follows proper grammar and structure; and (5) *Fluency*: whether the explanation flows smoothly with well-connected ideas. Each criterion is rated on a 5-point Likert scale. For detailed prompt, refer to Appendix.

As shown in Table 4, our P2C reasoning achieves the highest scores across all five criteria. This suggests that the structured perception-to-cognition decomposition produces more thorough and well-grounded explanations compared to the describe-

Table 4: Explanation quality evaluated by GPT-4o on 100 MHClip-Y samples (5-point Likert). [Mock data.]

Method	Info.	Sou.	Per.	Rea.	Flu.	Avg.
HVGuard	3.42	3.51	3.28	4.15	4.08	3.69
RAMF	3.38	3.45	3.35	4.02	3.95	3.63
Ours	**4.12**	**4.05**	**3.92**	**4.28**	**4.21**	**4.12**

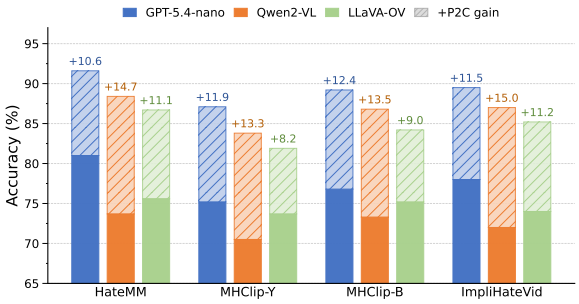

Figure 2: Generalizability of P2C CoT across different MLLMs. Applying our CoT design consistently improves performance over each MLLM's default prompting across all three datasets. [Mock data — to be replaced with real experiments.]

analyze-judge paradigm.

Generalizability across MLLMs. To assess whether our CoT design transfers across different MLLMs, we apply the P2C prompt to three models of varying scale: GPT-5.4-nano, Qwen2-VL (Wang et al., 2024b), and LLaVA-OV (Li et al., 2024). As shown in Figure 2, applying our P2C CoT consistently improves each MLLM's performance . This confirms that the perception-to-cognition decomposition is a generalizable design principle.

3.5 Interpolator Analysis

Correcting Hard Samples. To validate that the Whitened Neighbor Interpolator addresses the decision challenge on ambiguous content, we analyze *where* retrieval helps by binning test samples into quartiles based on the classifier's prediction confidence. As shown in Figure 3, retrieval gains are overwhelmingly concentrated in low-confidence bins. This pattern is consistent across all three datasets, confirming that retrieval acts as targeted decision support for ambiguous cases rather than a uniform post-hoc adjustment.

Enhancement of Transferability. Hateful video detection models are often trained on one platform but deployed on another. To evaluate cross-domain robustness, we conduct exhaustive transfer experiments: for each of the three datasets, we train

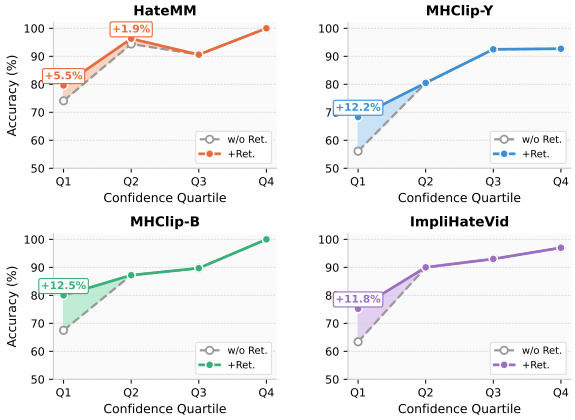

Figure 3: Retrieval gain by classifier confidence quartile across three datasets. Gains are concentrated in low-confidence bins, confirming that retrieval acts as targeted decision support for uncertain samples rather than a uniform calibration effect.

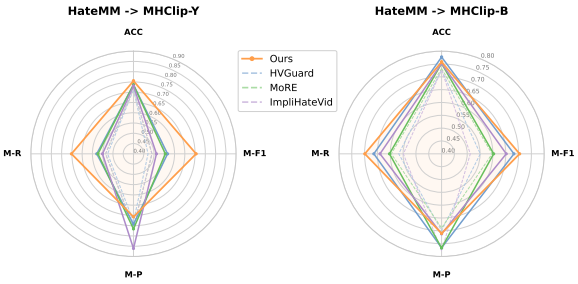

Figure 4: Cross-dataset transferability comparison. Radar plots show ACC, M-F1, M-P, M-R for each of six transfer directions. HATELENS outperforms HVGuard and ImpliHateVid in 4 out of 6 directions, with particularly strong gains on MHClip→HateMM transfers.

on one (source) and directly evaluate on the test set of another (target). We mainly focus on the performance of the baselines after plug-and-play integration of our interpolator module, where training instances from the source domain are used as the neighbor bank.

Figure 4 shows the transfer results for all six source→target directions. We observe that after applying our interpolator module, the performance of all baselines improves substantially. This provides strong evidence that the interpolator can serve as an effective test-time adaptation module across varied contexts. Furthermore, when combined with P2C reasoning, our method achieves the strongest performance, benefiting from the complementary strengths of both components and yielding a more transferable decision boundary.

[PLACEHOLDER: Qualitative Case Study]
Two examples showing: evidence ledger, relational meaning, alternative appraisals, top-3 nearest neighbors, and how retrieval corrects the classifier's prediction.

Figure 5: Case study. To be replaced.

3.6 Case Study

Figure 5 presents the predictions for two representative cases from the baselines and our method. In the first case, the hateful expression is relatively implicit. As a result, the other three baselines make incorrect predictions and fail to identify it as hateful. Methods without MLLMs do not provide explanations, which further increases decision difficulty, while MLLM-based methods remain at the level of surface multimodal analysis and produce judgments without collecting intermediate cognitive evidence. In contrast, our method fully exploits the reasoning capability of the MLLM, progressing from perception to cognitive evidence and finally to judgment, thereby successfully detecting this case. The second case is even more challenging, and all methods, including ours, initially classify it as non-hate. However, because our method is equipped with the interpolator with external retrieval at inference time, it can effectively leverage similar samples with annotated ground truth and ultimately correct the prediction.

4 Related Work

Our work is closely related to video-based hate speech detection. Early hate speech detection research mainly focused on text-based settings, typically using traditional machine learning classifiers (Davidson et al., 2017). With the rise of deep learning, increasing attention has been devoted to hate detection in multimodal settings, from the introduction of image-text meme benchmarks (Kiela et al., 2020) to more recent studies on hateful video detection (Das et al., 2023; Wang et al., 2024a; Rehman et al., 2025). For videos, most existing work extends conventional visual-textual feature fusion to three modalities, i.e., visual, textual, and audio fusion, through different fusion mechanisms (Céspedes-Sarrias et al., 2025; Zhang et al., 2024; Lang et al., 2025; Yue et al., 2025; Rehman et al., 2025). With the emergence of MLLMs, some studies have begun to explore different prompting techniques, either by generating rationales to assist the training of downstream conventional models (Jing et al., 2025; Cao et al., 2023), or by directly performing zero-shot inference with MLLMs (Yang et al., 2025). However, most existing methods share two common limitations: they do not provide a well-grounded understanding of video content, but mainly model *what* is in the video, and they lack an effective mechanism for handling ambiguous cases involving implicit and diverse hateful expression. In this work, we first propose to internally understand videos through a perception-to-cognition reasoning process, and then resolve ambiguity by consulting annotated precedents that provide meaningful local structural evidence.

5 Conclusion

We presented HATELENS, a framework that reframes hateful video detection as an appraisal problem. By grounding LLM-based analysis in Cognitive Appraisal Theory, our dual-stage pipeline produces structured intermediate representations—relational meaning and alternative appraisals—that capture the interpretive judgments necessary for distinguishing hateful endorsement from reporting, satire, and counterspeech. A case-based retrieval module further refines uncertain predictions by consulting similar labeled instances, providing targeted decision support for ambiguous content.

Beyond overall accuracy improvements, our most consequential practical result is the substantial reduction in false positives on benign content containing hate-adjacent language: HATELENS reduces FPR by 50–60% on quoted hate and counterspeech compared to existing methods. This has direct implications for content moderation, where false positives on legitimate speech are costly. We believe the appraisal-centered perspective offers a

productive direction for building more nuanced and trustworthy hate detection systems.

Limitations

Our framework relies on a commercial LLM (GPT-4.1-nano) for the appraisal stage, which introduces API costs and latency. The two-call pipeline approximately doubles inference time compared to single-call approaches. Additionally, our case-based retrieval module requires storing training set embeddings, though the storage overhead is minimal (<1MB for all datasets). The framework has been evaluated on English and Chinese datasets; its effectiveness on other languages and cultural contexts remains to be verified.

Ethics Statement

This work involves detecting hateful content in online videos, which necessarily requires engaging with offensive material. All datasets used are publicly available and were collected with appropriate ethical oversight by their original creators. We note several ethical considerations: (1) Hate speech annotations are inherently subjective and may reflect annotator biases; our method inherits these biases through training data. (2) Automated detection systems risk both false positives (censoring legitimate speech) and false negatives (failing to flag harmful content), with disparate impacts across communities. (3) Our system should be used as a *decision-support tool* alongside human review, not as an autonomous moderator. (4) The LLM component may generate harmful text during analysis; we process outputs programmatically without displaying them to end users. (5) Cross-cultural hate norms vary; performance on Chinese content may not reflect all cultural contexts within Chinese-speaking communities.

References

Rui Cao, Roy Ka-Wei Lee, Wen-Haw Chong, and Jing Jiang. 2023. Pro-cap: Leveraging a frozen vision-language model for hateful meme detection. In *ACM International Conference on Multimedia*.

Berta Céspedes-Sarrias and 1 others. 2025. Mm-hsd: Multi-modal hate speech detection in videos.

Sanyuan Chen and 1 others. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. volume 16, pages 1505–1518.

Christopher Clarke, Matthew Hall, Jeff Sheridan, James Fiacco, and Bing Li. 2023. Rule by example: Harnessing logical rules for explainable hate speech detection. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 364–378.

Mithun Das, Rohit Raj, Punyajoy Saha, Binny Mathew, Manish Gupta, and Animesh Mukherjee. 2023. Hatemm: A multi-modal dataset for hate video classification. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 17, pages 1014–1023.

Thomas Davidson, Dana Warmsley, Michael Macy, and Ingmar Weber. 2017. Automated hate speech detection and the problem of offensive language. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 11, pages 512–515.

Steven B. Davis and Paul Mermelstein. 1980. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 28(4):357–366.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186.

Alexey Dosovitskiy and 1 others. 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.

Yiheng Jing, Mingming Zhang, Yong Zhuang, Jiacheng Guo, Juan Wang, Xiaoyang Xu, Wenzhe Yi, Keyan Guo, and Hongxin Hu. 2025. Hvguard: Utilizing multimodal large language models for hateful video detection. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 8993–9006.

Douwe Kiela and 1 others. 2020. The hateful memes challenge: Detecting hate speech in multimodal memes. In *Advances in Neural Information Processing Systems*, volume 33.

Kate Klonick. 2018. The new governors: The people, rules, and processes governing online speech. *Harvard Law Review*, 131:1598.

Jian Lang, Rongpei Hong, Jin Xu, Yili Li, Xovee Xu, and Fan Zhou. 2025. Biting off more than you can detect: Retrieval-augmented multimodal experts for short video hate detection. In *Proceedings of the ACM Web Conference 2025*, pages 2763–2771.

Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2024. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*.

Mohammad Zia Ur Rehman, Anukriti Bhatnagar, Omkar Kabde, Shubhi Bansal, and Nagendra Kumar. 2025. Implihatevid: A benchmark dataset and two-stage contrastive learning framework for implicit hate speech detection in videos. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 17209–17221.

Hayato Tsukagoshi and Ryohei Sasano. 2025. Redundancy, isotropy, and intrinsic dimensionality of prompt-based text embeddings. In *Findings of the Association for Computational Linguistics: ACL*.

Han Wang, Tan Nurbek, and Ee-Peng Lim. 2024a. Multihateclip: A multilingual benchmark dataset for hateful video detection on youtube and bilibili. In *Proceedings of the ACM International Conference on Multimedia*.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, and 1 others. 2024b. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*.

Shuonan Yang, Tailin Chen, Jiangbei Yue, Guangliang Cheng, Jianbo Jiao, and Zeyu Fu. 2025. Reasoning-aware multimodal fusion for hateful video detection. *arXiv preprint arXiv:2512.02743*.

Shuonan Yang, Yuchen Zhang, and Zeyu Fu. 2026. Training-free and interpretable hateful video detection via multi-stage adversarial reasoning. *arXiv preprint arXiv:2601.15115*.

Lingjie Yi and 1 others. 2025. Geometry of long-tailed representation learning: Rebalancing features for skewed distributions. In *International Conference on Learning Representations*.

Jiangbei Yue and 1 others. 2025. Multihategnn for hateful video classification. In *British Machine Vision Conference*.

Yinghui Zhang, Tailin Chen, Yuchen Zhang, and Zeyu Fu. 2024. Enhanced multimodal hate video detection via channel-wise and modality-wise fusion. In *2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 183–190.

A Implementation Details

LLM Configuration. We use GPT-4.1-nano as the backbone LLM for both stages of our pipeline. For Stage 1 (evidence extraction), the model receives a 2×2 quad image of representative frames along with the video title and transcript. For Stage 2 (appraisal inference), the model receives only the textual evidence ledger from Stage 1. Both stages use structured JSON output format. The average cost per video is approximately $0.002 for the two-call pipeline.